AI SERVICES IN HEALTHCARE

Hospital Efficiency & Clinical Accuracy

Case Study Report

Services Covered:

1. **Sepsis Detection & Early Alert System**
2. **Nurse Shift Handoff Documentation Assistant**

SERVICE 1

Sepsis Detection & Early Alert System

1. Problem Statement

Sepsis is one of the leading causes of preventable in-hospital mortality. Its early-stage indicators — mildly elevated heart rate, subtle temperature shifts, marginally abnormal lab values — are easy to overlook individually, particularly on busy wards where a single nurse may be responsible for six or more patients simultaneously.

Current detection relies on reactive, manual observation. By the time a clinician identifies a clear cluster of sepsis indicators, the patient may already be progressing toward septic shock, where treatment efficacy drops significantly with every passing hour. The core problem is not a lack of data — hospitals collect vitals and lab results continuously — but a lack of a system that monitors all patients at once, connects subtle signals across time, and raises an alert before the condition becomes clinically obvious.

Core Impact: Each hour of delayed sepsis treatment increases mortality risk by approximately 7%. For a mid-sized hospital, even a modest improvement in detection speed translates directly to lives saved and ICU admissions avoided.

2. Solution Parameters

The AI service operates as a passive, real-time monitoring layer that runs continuously in the background across all admitted patients. It does not replace clinical judgment — it surfaces risk so that clinicians can act faster.

Core Functionality

- Continuously ingests structured patient data: vitals (HR, BP, temperature, respiratory rate, SpO2), lab results (WBC, lactate, creatinine, CRP), and nursing observation notes.
- GPT-4o analyzes multi-parameter combinations over time windows (e.g., last 2, 6, and 12 hours) to detect sepsis-consistent patterns aligned with SOFA/qSOFA criteria.
- Generates tiered alert levels: Watch (early warning), Concern (prompt review recommended), and Critical (immediate intervention flagged).
- Each alert includes a brief AI-generated clinical rationale, listing the specific parameters that triggered the flag, making it actionable rather than just alarming.
- Alert delivery via API push to nursing dashboard, EMR notification, or SMS/pager depending on severity tier.

Prompt Engineering Approach

- System prompt encodes SOFA scoring criteria, qSOFA thresholds, and hospital-specific escalation logic.
- Patient data is formatted as structured context per request (rolling time-window snapshots).
- Output is strictly formatted JSON: { risk_level, confidence, triggered_indicators, rationale, recommended_action }.
- Prompt includes few-shot examples of true-positive and false-positive cases to calibrate sensitivity vs. specificity.

Constraints & Safeguards

- Alert fatigue mitigation: alerts are suppressed if the same patient was flagged within the last 4 hours unless a new critical parameter threshold is crossed.
- All AI output is advisory only — the system is positioned as a decision-support tool, not a diagnostic one.
- Audit log of all alerts generated and clinician acknowledgment timestamps for compliance and retrospective review.

3. Measures of Success

The following KPIs are recommended for evaluating the proof of concept during a controlled ward pilot (target: 90-day period).

Component	Technology / Tool	Purpose
Time-to-Detection	Avg. time from sepsis onset to alert	Target: reduce by 30-50% vs. baseline
Alert Precision	True positive rate of Critical/Concern alerts	Target: >75% precision to avoid fatigue
False Positive Rate	Alerts that did not correlate with sepsis	Target: <25% across all tiers
Mortality Rate (pilot ward)	Sepsis-related mortality during pilot period	Compare to 6-month pre-pilot baseline
ICU Escalation Rate	Patients escalated to ICU from sepsis	Directional reduction expected
Nurse Acknowledgment Rate	% of alerts reviewed within 15 minutes	Target: >85% (proxy for trust/usability)

4. Architecture

Component	Technology / Tool	Purpose
Data Ingestion	Python (HL7/FHIR parser or CSV feed)	Pulls vitals & labs from EMR/HIS at configurable intervals (e.g., every 5–15 min)
AI Engine	OpenAI GPT-4o API	Multi-parameter risk analysis, rationale generation, alert classification
Backend API	FastAPI (Python)	Exposes endpoints: /analyze-patient, /get-alerts, /acknowledge-alert
Prompt Layer	Python prompt templates + LangChain (optional)	Manages structured patient context injection and output parsing
Data Store	PostgreSQL	Stores patient snapshots, alert history, and acknowledgment logs
Alert Delivery	FastAPI webhooks / REST push	Delivers structured alert payloads to nursing dashboard or EMR integration
Testing	Postman collections	Functional API testing, alert payload validation, edge case simulation

Data Flow (Simplified)

- Step 1 — EMR/HIS exports patient vitals and lab data at set intervals to a secure ingestion endpoint.
- Step 2 — Python service formats each patient's rolling data window into a structured prompt payload.
- Step 3 — GPT-4o API call returns a JSON alert object (risk_level, indicators, rationale).
- Step 4 — FastAPI processes the response, applies alert fatigue suppression logic, and persists to database.
- Step 5 — Alert is pushed to the nurse dashboard or EMR notification system based on severity tier.
- Step 6 — Nurse acknowledges or escalates; timestamp is logged for audit and retrospective KPI tracking.

5. Proposed Timeline (Proof of Concept)

Target scope: One ward (20-30 patient beds). Total PoC duration: 3-5 weeks.

SERVICE 2

Nurse Shift Handoff Documentation Assistant

1. Problem Statement

Shift handoffs are consistently identified as one of the highest-risk communication events in hospital care. At the end of every shift, nurses are required to verbally and/or documentarily brief the incoming nurse on the status of every patient under their care — medications administered, condition changes, pending tasks, outstanding doctor orders, and issues to watch for.

In practice, this process is rushed, unstandardized, and heavily reliant on individual memory. There is no enforced format in most wards. The outgoing nurse is fatigued and time-pressured; the incoming nurse may be absorbing information on six or more patients simultaneously. The result is that critical details are routinely missed or inconsistently communicated — a skipped dose, an unreported symptom, a pending lab result that goes unnoticed until the next shift.

Core Impact: Hospitals are liable for any mistakes their nurses make, and nurses work long and grueling hours increasing these chances of fatal mistakes, this service aims to limit and eventually get rid of this liability for hospitals and ease the stress of nurses.

2. Solution Parameters

The AI assistant acts as an always-available documentation layer that collects, structures, and synthesizes patient information throughout the shift, producing a consistent, prioritized handoff summary at the end of each shift — automatically.

Core Functionality

- At shift start, the system loads each patient's current chart data from the EMR: active medications, last vitals, pending orders, diagnosis, and care plan.
- Throughout the shift, nurses can append quick freeform notes via a simple text or voice input (e.g., 'Patient reported chest tightness at 14:30, doctor notified. Awaiting ECG order.').
- At shift end, GPT-4o synthesizes the structured chart data and the nurse's freeform notes into a clean, standardized handoff summary per patient.
- Summary sections follow SBAR format (Situation, Background, Assessment, Recommendation) — a widely accepted clinical communication standard.
- Patients are automatically ranked by priority in the handoff document: critical flags and pending actions appear first.

- Output is delivered as a formatted document viewable on ward terminals or mobile devices, and optionally printed or sent to the EMR.

Prompt Engineering Approach

- System prompt instructs the model to strictly follow SBAR output structure and clinical tone, with no unnecessary preamble.
- Patient context is injected in a structured schema: { patient_id, diagnosis, medications, last_vitals, pending_orders, nurse_notes }.
- Priority ranking logic is embedded in the prompt: flag any patient with a pending critical order, abnormal vitals in last 2 hours, or nurse-noted concern.
- Model is instructed to never infer or assume clinical details not present in the provided data — output must be grounded strictly in the input.
- Fine-tuning (Phase 2 – after deployment): a dataset of high-quality, hospital-approved handoff summaries is used to further align output style to institutional standards.

Constraints & Safeguards

- The AI produces a draft summary — the outgoing nurse reviews and confirms before handoff. No fully automated release.
- All source data references are preserved in the output so incoming nurses can trace any item back to the raw chart entry.
- Voice input is transcribed using OpenAI Whisper API for accuracy, with the raw transcript stored alongside the processed note for audit.

3. Measures of Success

Recommended KPIs for a 90-day controlled pilot on one ward (2 shifts per day, approx. 10 nurses in rotation).

Component	Technology / Tool	Purpose
Handoff Completion Time	Time spent per shift on handoff documentation	Target: reduce by 30-40% vs. manual baseline
Post-Handoff Error Rate	Medication errors & missed follow-ups in 24h after handoff	Target: measurable reduction vs. 3-month pre-pilot baseline
Summary Completeness Score	% of required SBAR fields populated per summary	Target: >95% completeness on AI-generated summaries
Nurse Adoption Rate	% of eligible shifts using the AI assistant vs. manual	Target: >80% by end of pilot Week 6
Edit Rate	% of AI summaries that nurses modified before sign-off	Diagnostic metric — high edit rate indicates prompt tuning needed

Nurse Satisfaction Score	Post-shift survey rating (1–5 scale)	Target: avg. score >3.8 for usability and trust

4. Architecture

Component	Technology / Tool	Purpose
Data Ingestion	Python (FHIR REST API or CSV/HL7 adapter)	Pulls patient chart data from EMR at shift start and on-demand refresh
Voice Input (optional)	OpenAI Whisper API	Transcribes nurse voice notes to text; stored as raw transcript
AI Engine	OpenAI GPT-4o API	Synthesizes chart data + nurse notes into structured SBAR handoff summary
Backend API	FastAPI (Python)	Endpoints: /load-patients, /add-note, /generate-summary, /confirm-handoff
Prompt Layer	Python prompt templates	SBAR structure enforcement, priority ranking logic, data grounding rules
Data Store	PostgreSQL	Stores nurse notes, generated summaries, confirmation logs, and audit trail
Frontend (PoC)	Simple React web app or ward terminal UI	Nurse-facing interface for note entry and summary review/confirmation
Testing	Postman collections	API endpoint testing, summary generation validation, edge case inputs

Data Flow (Simplified)

- Step 1 — At shift start, system fetches current patient chart data from EMR for all patients assigned to the nurse.
- Step 2 — During the shift, nurse submits freeform text or voice notes via the web interface. Notes are stored and timestamped.
- Step 3 — At shift end, nurse triggers summary generation. FastAPI compiles chart data + nurse notes into a structured prompt payload.
- Step 4 — GPT-4o returns a formatted SBAR summary per patient with priority ranking applied.
- Step 5 — Nurse reviews the draft summary on screen, makes any edits, and confirms.

- Step 6 — Confirmed summary is stored in the database, optionally pushed to the EMR, and made available to the incoming nurse on their terminal.

5. Proposed Timeline (Proof of Concept)

Target scope: One ward, two nursing shifts per day, approx. 10 nurses in rotation. Total PoC duration: 5-8 weeks.

End of Case Study Document